Expressions and equations

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 848 to the nearest hundred.

2. Round 8.72 to the nearest whole number.

3. Use 20×5 to estimate 19.8×5.1.

4. Use $60 \div 3$ to estimate $59.4 \div 3.1$.

5. Solve $1.5x = 4.80$.

6. Expand $4(x + 5)$.

7. Solve $7x = 56$.

8. Write a story for $\frac{5}{6} \div \frac{1}{3}$. Solve it.

9. Find $9.6 \div 8$.

10. Find $2.8 \div 0.07$.

11. A ball costs \$4.65 and a card costs \$2.78. Find the change from \$10.

12. Find $10 - 1.286$.

13. Each pack has b beads. Write the total in 6 packs. Find it when $b = 4$.

14. Omar solves $6x = 42$ by writing $x = 42 - 6 = 36$. Explain and repair his work.

15. A packet contains 8 cards. Write the number in p packets plus 3 loose cards; evaluate at $p = 7$.

16. Evaluate $3t + 8$ when $t = 7$. Show the multiplication first.

17. Evaluate $3^2 + 8$. Explain what the exponent means.

18. Expand $3(x + 8)$ and check at $x = 7$.

CHALLENGE / Explain your plan and check the result

C1. A shop has a \$45.75 budget for identical folders and a \$9 delivery charge. Seven folders fit the budget exactly. Find the folder price. Can the shop buy 9 folders if delivery becomes free? Show the remaining amount.

C2. A square tile has side x cm. A border strip adds a rectangle 2 cm wide along one side. Write the total area. Compare $x = 1$, $x = 3$, and $x = 5$. Do the two increases of 2 in x produce equal increases in area?